Superdeterminism lives below every resolution; a floor gives it a lifetime

A connected mixing Anosov setting map and the stated uniformly resolved conditional preparation laws give an exponential bound on setting/hidden-data covariance. Deterministic dynamics alone supplies neither statistical independence nor its failure. The preparation resolution δ below is an additional premise, not a derived action floor. For a preparation whose device part is resolved at scale δ, the covariance between any bounded hidden variable f and a setting chosen n steps later obeys

$$|\mathrm{Cov}_\rho(f, g\circ T^n)| \le 2C_T\,K\,\sup|f|\,\|g\|_{\mathrm{Lip}}\,\delta^{-(d+1)}\theta^n,$$

so the correlation is below ϵ after

$$n_*(\delta) = \frac{(d+1)\ln(1/\delta) + \ln\!\big(2C_T K \sup|f|\,\|g\|_{\mathrm{Lip}}/\epsilon\big)}{\ln(1/\theta)}$$

steps. Classical mechanics allows every $\delta > 0$, and $n_* \to \infty$ as $\delta \to 0$: a conspiracy can be stored in ever finer structure for as long as needed. If a floor h on phase-space area per canonical pair constrains the prepared state, then $\ln(1/\delta) = \frac{1}{2}\ln(A_0/h)$ with A_0 the device's action scale per pair, and the conspiracy has a finite lifetime $n_* \approx \frac{d+1}{2\ln(1/\theta)}\ln(A_0/h)$, the logarithmic structure of the Ehrenfest time of quantum chaos (Berman and Zaslavsky 1978; metadata). The dependence of hidden-variable events on setting events then decays exponentially after an onset of order $2n_*$ (Remark in §1), and relaxed Bell inequalities bound local deterministic models that respect the floor, under the hypotheses stated there. The limit $h \to 0$ that Newton takes is exactly what reopens the loophole.

The user's remark that prompted this (2026-09-28/29): classical mechanics is accidentally superdeterministic too, and the programme holds that it has a mistake somewhere. The link is that typicality rests on the Liouville measure, whose unit is an action per degree of freedom, the same object whose unlimited refinement the fifth-postulate note identifies as Newton's joint determinacy.

1. Setting and the proposition

The phase space is a product $M_{\mathrm{sys}} \times M$: the measured system, a probability space with reference measure ν and otherwise arbitrary, and the setting device, a compact **connected** d-dimensional manifold with normalized volume μ. The dynamics is $\mathrm{id} \times T$, with T a C^2 volume-preserving Anosov diffeomorphism; only the device needs chaos. By Anosov's theorem such a T is ergodic for μ, hence transitive, hence topologically mixing because M is connected, and μ is its equilibrium state. Bowen's theory then gives exponential decay of correlations for Hölder, hence Lipschitz, observables (Bowen 1975; metadata, theorem numbers still to be checked at passage level): with $\|F\|_{\mathrm{Lip}} = \sup|F| + \mathrm{Lip}(F)$,

$$\Big|\int F\,(G\circ T^n)\,d\mu - \int F\,d\mu \int G\,d\mu\Big| \le C_T \|F\|_{\mathrm{Lip}}\|G\|_{\mathrm{Lip}}\,\theta^n, \qquad \theta < 1,$$

where θ is the transfer-operator gap rather than a Lyapunov exponent.

A preparation is a probability density $\rho(x_1, x)$ on $M_{\mathrm{sys}} \times M$ with respect to $\nu \otimes \mu$. The hidden variable is $\lambda = f(x_1)$ with f merely bounded and measurable, and the setting is $s = g(T^n x)$ with g Lipschitz; λ is fixed at time 0 and the setting is read n steps later. The preparation is **resolved at scale** δ in the device if $\int \|\rho(x_1,\cdot)\|_{\mathrm{Lip}}\,d\nu(x_1) \le K\delta^{-(d+1)}$, i.e. the device's density conditional on the system is Lipschitz at scale δ on average. For example, a conditional density smoothed by a mollifier ϕ_δ of width $\delta \le 1$ has height at most $\delta^{-d}\sup\phi$ and slope at most $\delta^{-(d+1)}\mathrm{Lip}\,\phi$, so $K = \|\phi\|_{\mathrm{Lip}}$.

Proposition. For a preparation resolved at scale δ in the device, $|\mathrm{Cov}_\rho(f, g \circ T^n)| \le 2C_T K \sup|f|\,\|g\|_{\mathrm{Lip}}\delta^{-(d+1)}\theta^n$.

Proof. For each x_1 apply the decay estimate with $F = \rho(x_1,\cdot)$: $\int \rho(x_1,x)g(T^n x)\,d\mu = \rho_1(x_1)\int g\,d\mu + E(x_1)$, with ρ_1 the system marginal and $|E(x_1)| \le C_T\|\rho(x_1,\cdot)\|_{\mathrm{Lip}}\|g\|_{\mathrm{Lip}}\theta^n$. Then $E_\rho[f\,g\circ T^n] = E_\rho[f]\int g + \int fE\,d\nu$ and $E_\rho[g\circ T^n] = \int g + \int E\,d\nu$, so $\mathrm{Cov}_\rho = \int fE\,d\nu - E_\rho[f]\int E\,d\nu$, bounded by $2\sup|f|\int|E|\,d\nu$. □

The product form frees the hidden variable from any regularity: a superdeterminist may choose f at will, and the conspiracy must then sit in the device's conditional density.

Remark (from covariances to measurement dependence). Bell's theorem uses the laws of λ conditional on the settings. Hall measures their dependence by the L^1 (total-variation) distance between these conditional laws and gives relaxed Bell inequalities in its terms, together with a deterministic model of the singlet that needs only a partial dependence (Hall 2010); Barrett and Gisin quantify the missing independence in information-theoretic form (Barrett and Gisin 2011). Both: Crossref metadata verified 2026-09-29, content from memory. Let a setting event $A \subset M$ have a piecewise smooth boundary whose w-neighbourhood has μ-measure at most $c_A w$, and let B be any measurable hidden-variable event. Replace 1_A by a function with values in $[0,1]$ and Lipschitz constant $1/w$ that differs from it only in that neighbourhood, so its norm is at most $2/w$ for $w \le 1$. The device marginal has $\sup\rho_{\mathrm{dev}} \le K\delta^{-(d+1)}$, and since T preserves μ the layer costs at most $K\delta^{-(d+1)}c_A w$ in $E_\rho[1_B\,1_A\circ T^n]$ and in the product of the means. Hence

$$\sup_B |\mathrm{Cov}_\rho(1_B, 1_A\circ T^n)| \le K\delta^{-(d+1)}\big[4C_T\theta^n w^{-1} + 2c_A w\big],$$

and $w = \theta^{n/2}$ gives $\varepsilon_n = K\delta^{-(d+1)}(4C_T + 2c_A)\theta^{n/2}$. For setting cells A_j with probabilities $p_j = P_\rho(T^n x \in A_j)$, which tend to $\mu(A_j)$ at the same rate, the conditional laws of λ satisfy $\sup_B |P(B \mid A_j) - P(B \mid A_k)| \le \varepsilon_n(1/p_j + 1/p_k)$. The onset for events is twice the Lipschitz n_*, with the same logarithm of A_0/h. The conclusion that relaxed Bell inequalities then exclude the singlet correlations holds under three stated hypotheses: (a) the delay n between fixing λ and reading the settings exceeds this onset; (b) setting cells have regular boundaries (hidden-variable events need none); (c) both wings' settings come from one Anosov device, or from two, whose product is again Anosov on a connected manifold.

Scope of the floor. The floor is imposed on the preparation; the dynamics is Newton's. The pushforward of ρ stretches along unstable directions, so its Lipschitz norm grows like θ^{-n} and violates the floor at about the same n_*. A floor enforced at every step, by a coarse-graining map, could only add decorrelation; that is left unproved. Read quantum mechanically, Husimi functions are resolved at $\delta = (\hbar/A_0)^{1/2}$ in this sense, but classical evolution of observables tracks the quantum one only up to the Ehrenfest time, i.e. up to n_*. The floor meant here is the programme's classical floor, and the Ehrenfest match is formal. The translation $\ln(1/\delta) = \frac{1}{2}\ln(A_0/h)$ assumes the cell is split isotropically between q and p after both are normalized by $A_0^{1/2}$, with $d = 2N$ for N canonical pairs.

2. What it says, and about whom

- **Classical mechanics.** The laws leave δ free. For a chaotic setting device, independence is therefore an assumption about the data, and the superdeterministic option is a device ensemble with structure at the scale θ^n or finer along the stable directions, correlated with λ. This is the precise sense of "accidentally superdeterministic"; for an integrable device, coarse data can carry a correlation forever, and the statement is proved only in the chaotic case.
- **A floor.** The proposition needs the floor to constrain the prepared physical state (its phase-space density), beyond constraining records. Under that premise, correlations of a chaotic setting device with a hidden variable decay after a time logarithmic in A_0/h, the Ehrenfest-type scale at which classical and quantum chaos part.

- **'t Hooft's automaton** lies outside the proposition. Its ontic states are exact cells and its correlations are carried by exact labels, which carry no metric a floor could smooth; the automaton has its own floor, the cell, and a permutation of a finite set is periodic and never mixes, so a lifetime statement for it would need an infinite automaton (discrete-substrate note).
- **Palmer's invariant set postulate** places states on a fractal, measure-zero invariant set and counterfactual settings off it (Palmer 2009; read at the level of its title and our memory of the framework). A Lipschitz density cannot live on a measure-zero set, so for that formulation the two premises are in direct conflict. Later versions of invariant set theory, as we recall them, introduce a finest fractal scale; there the comparison becomes quantitative, that scale against A_0/h. The superdeterminist position is surveyed in Hossenfelder and Palmer (2020) (metadata).
- **Quantum mechanics** is untouched: its Bell violations come from entangled states, and it assumes independent settings, consistently with the proposition.

3. Consequence for STATE

A small addition to the Newton side: for chaotic setting devices a floor on the prepared state makes Bell's measurement dependence decay exponentially after an Ehrenfest-type onset, and the continuum limit $h \to 0$ reopens the loophole. It is a statement about what the floor buys, with the floor assumed; the necessity question is unchanged. The connectedness, mixing, conditional product formulation and hypotheses (a)–(c) are essential. Written model checks were internal, not independent peer review.